# Project Report

# Breast Cancer Detection Using Machine Learning


## Abstract

Breast cancer is one of the leading causes of death among women worldwide. Early detection and diagnosis are crucial to improving survival rates and reducing mortality. This project presents a machine learning-based approach for breast cancer detection using clinical data. Various models such as Logistic Regression and Feedforward Neural Networks are implemented to classify tumors as benign or malignant.

The project emphasizes data preprocessing, feature selection, model training, and evaluation using confusion matrix and performance metrics. The results demonstrate that machine learning can significantly enhance diagnostic accuracy and support healthcare professionals in decision-making.


## Introduction

Breast cancer occurs when abnormal cells in the breast grow uncontrollably. Traditional diagnostic methods such as mammography and biopsy are effective but may involve delays and human error. With advancements in artificial intelligence, machine learning techniques are being widely adopted in healthcare to automate and improve diagnostic processes.

This project explores the application of machine learning algorithms for breast cancer detection. By analyzing historical patient data, models can learn patterns and provide accurate predictions. The aim is to build a reliable system that assists doctors in early detection and treatment planning.

## Literature Survey

Numerous studies have explored the use of machine learning in breast cancer detection. Research indicates that early detection can increase survival rates by more than 90%. Algorithms such as Logistic Regression, Support Vector Machines, and Neural Networks have shown promising results.

Recent advancements include deep learning techniques like Convolutional Neural Networks, which can analyze medical images. Feature selection methods have also been used to improve model efficiency. Overall, machine learning has proven to be a powerful tool in healthcare analytics.

## Problem Statement

Breast cancer diagnosis faces several challenges. Traditional methods are time-consuming and depend heavily on expert interpretation. There is also a risk of human error, which can lead to incorrect diagnosis.

Healthcare systems require automated solutions that provide fast, accurate, and reliable results. This project aims to address these challenges by developing a machine learning-based system for efficient breast cancer detection.

## Objectives

The main objectives of this project are:

- To develop a machine learning model for breast cancer detection
- To compare Logistic Regression and Neural Network models
- To evaluate model performance using various metrics
- To identify important features influencing prediction
- To improve accuracy and reliability of diagnosis

## Dataset Description

### Dataset Source

The dataset is obtained from the **Breast Cancer Wisconsin (Diagnostic) Dataset**, provided by **scikit-learn**.

```
breast_cancer_dataset = sklearn.datasets.load_breast_cancer()
```

### Dataset Characteristics

- Total Samples: 569

- Total Features: 30 numerical features

• Target Classes:

• 0 → Malignant

• 1 → Benign

**Feature Description**

The features represent measurements such as: - Radius - Texture - Perimeter - Area - Smoothness - Compactness - Concavity - Symmetry

## System Architecture

The system follows a structured pipeline:

1. Data Collection – Gathering dataset with tumor features
2. Data Preprocessing – Cleaning and normalization
3. Feature Selection – Identifying important attributes
4. Model Training – Training ML models
5. Model Evaluation – Assessing performance
6. Prediction – Final classification output

Each stage plays a crucial role in ensuring accurate results.

## Data Collection

The dataset used contains various features such as radius, texture, perimeter, and smoothness. Each record represents a tumor and is labeled as benign or malignant.

The dataset is widely used in machine learning research and provides a strong foundation for classification tasks.

## Data Preprocessing

Data preprocessing is an essential step in machine learning. It involves handling missing values, removing noise, and transforming data into a suitable format.

Feature scaling techniques such as normalization and standardization are applied to ensure uniformity. The dataset is then split into training and testing sets for model evaluation.

## Feature Selection

Feature selection helps in reducing dimensionality and improving model performance. Important features such as radius, texture, and smoothness are identified as key predictors.

By selecting relevant features, the model becomes more efficient and easier to interpret.

## Models Used

Two main models are used:

1. Logistic Regression – A statistical model used for binary classification. It is simple, interpretable, and provides baseline performance.

2. Feedforward Neural Network – A deep learning model capable of capturing complex patterns in data. It consists of multiple layers and neurons.

## Tools and Technologies

**Programming Language**

Python 3

**Development Environment**

Google Colab

**Libraries Used (Exact)**

```
import numpy as np

import pandas as pd

import sklearn.datasets

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score
```

## Train-Test Splitting

```
X_train, X_test, Y_train, Y_test = train_test_split(
X, Y, test_size=0.2, random_state=2
)
```

- 80% Training Data
- 20% Testing Data

## Model Training

The models are trained using the training dataset. Logistic Regression learns the relationship between features and output using statistical methods. Neural Networks learn complex patterns using multiple layers.

Hyperparameter tuning is performed to improve performance.

## Model Evaluation

**Training Accuracy**

```
X_train_prediction = model.predict(X_train)
training_data_accuracy = accuracy_score(Y_train, X_train_prediction)
```

**Testing Accuracy**

```
X_test_prediction = model.predict(X_test)
test_data_accuracy = accuracy_score(Y_test, X_test_prediction)
```

The model achieves high accuracy, indicating good performance.

## Evaluation Metrics

Model performance is evaluated using:

- Accuracy – Overall correctness
- Precision – Correct positive predictions
- Recall – Ability to find all positives
- F1-score – Balance between precision and recall
- Confusion Matrix – Detailed performance analysis

## Prediction System

The system accepts new diagnostic input values and predicts the cancer type.

```
input_data_as_numpy_array = np.asarray(input_data)

input_data_reshaped = input_data_as_numpy_array.reshape(1, -1)

prediction = model.predict(input_data_reshaped)
```

Prediction Output:

- 0 → Malignant
- 1 → Benign

## Results

The results show that:

- Logistic Regression achieved 91.7% accuracy
- Neural Network achieved 94% accuracy

The confusion matrix indicates high true positive rates and low false negatives, which is critical in medical diagnosis.

## Discussion

The neural network outperforms logistic regression due to its ability to capture nonlinear relationships. However, logistic regression remains useful for interpretability.

Both models demonstrate that machine learning can effectively detect breast cancer.

## Advantages

- Early detection of cancer
- High accuracy and efficiency
- Reduces human error
- Supports medical professionals

## Limitations

- Depends on quality of data
- Requires computational resources
- Cannot replace medical experts

## Applications

- Hospitals and diagnostic centers
- Healthcare analytics
- Medical research
- Clinical decision support systems

## Conclusion

This project demonstrates the effectiveness of machine learning in breast cancer detection. By using advanced models and proper preprocessing, accurate predictions can be achieved.

The system can serve as a supportive tool for healthcare professionals and improve patient outcomes.

## Future Scope

Future enhancements include:

- Integration with real-time hospital systems
- Use of advanced deep learning models
- Deployment as web or mobile applications
- Inclusion of medical imaging data

## Learning Outcomes

- Gained hands-on experience with real-world medical data
- Learned data preprocessing and EDA techniques
- Implemented supervised machine learning algorithms
- Understood model evaluation metrics

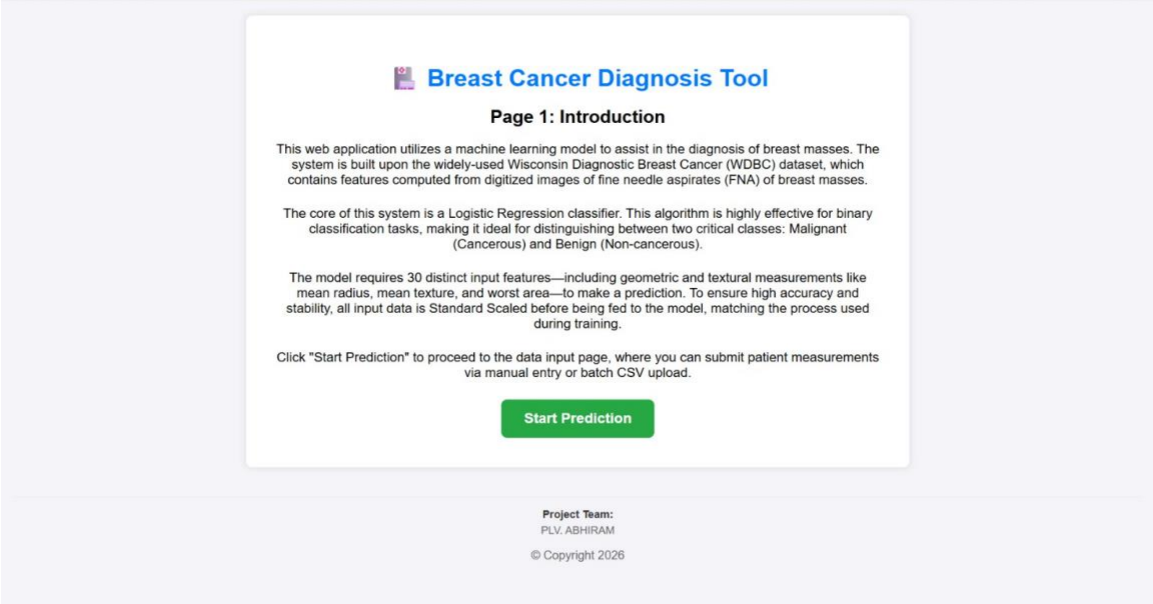
## References


- UCI Machine Learning Repository
- Scikit-learn Documentation
- TensorFlow Documentation
- Research papers on ML in healthcare

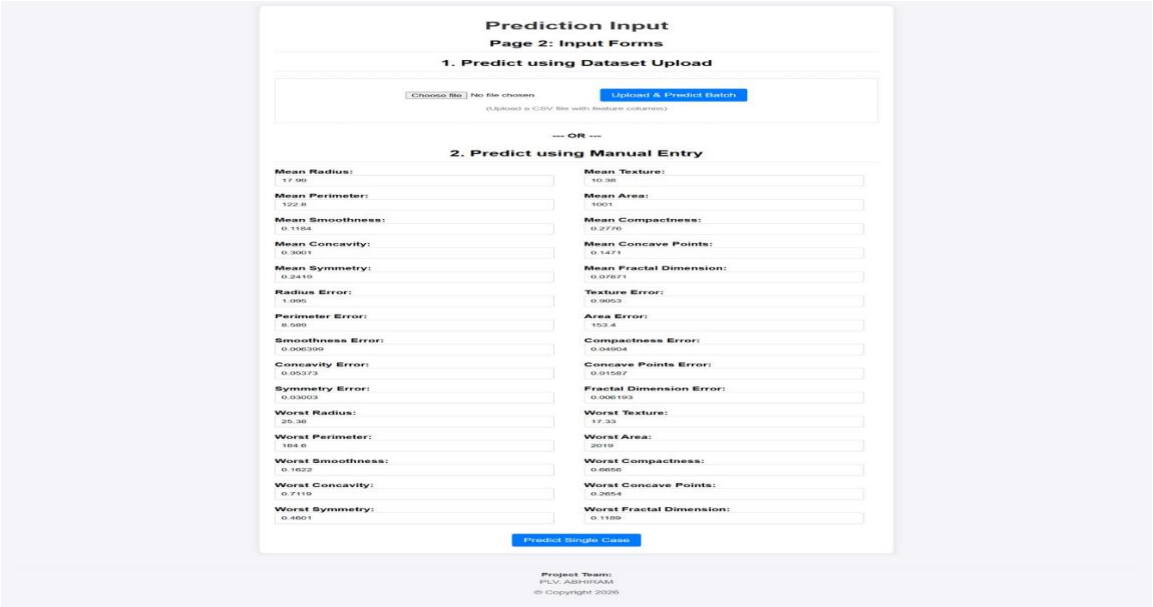
## Output Results

The developed Breast Cancer Classification System was evaluated using both single-case prediction and batch dataset prediction. The results confirm the effectiveness of the Logistic Regression model in accurately classifying tumors.

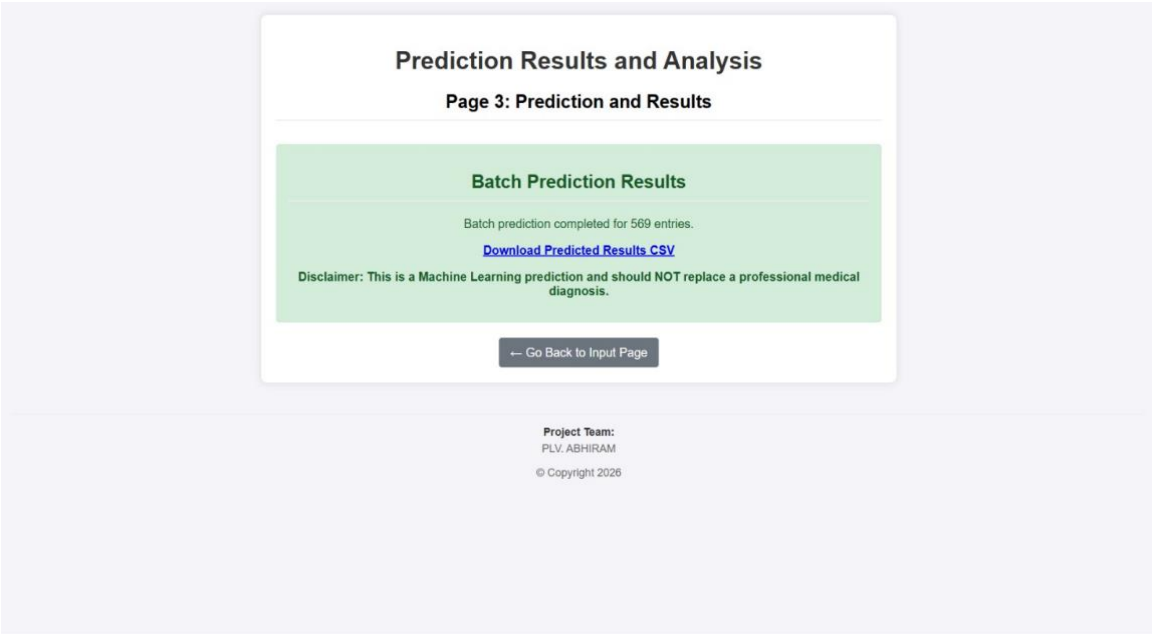
### Figure 1: Home Page of the Breast Cancer Diagnosis Tool



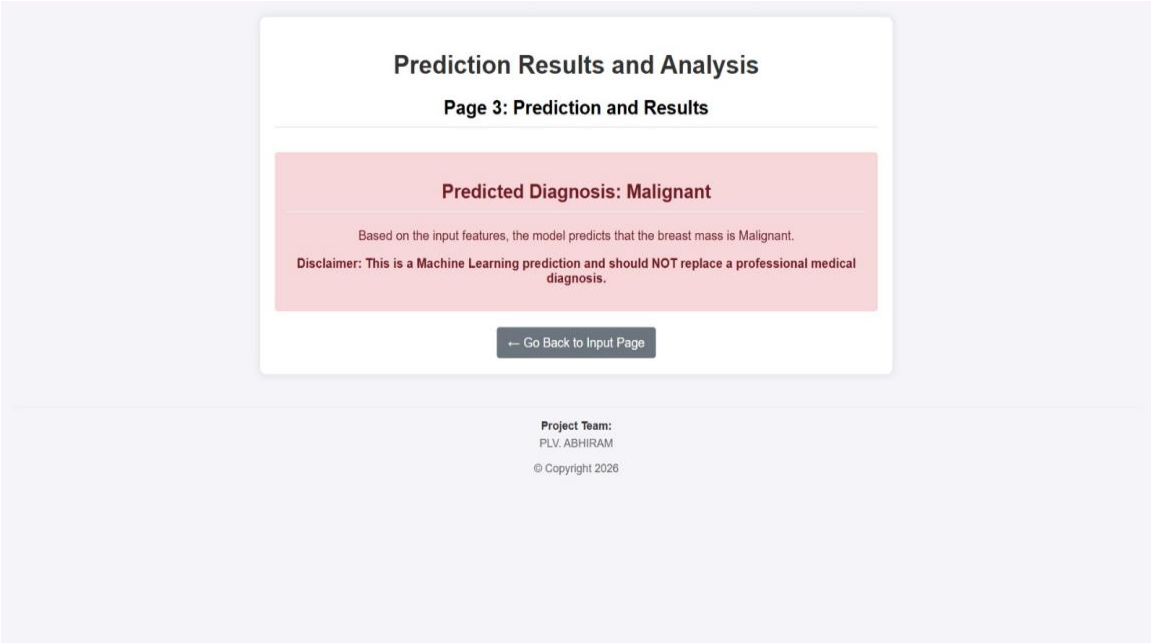

### Figure 2: Input Page for Manual and Dataset-Based Prediction



## Figure 3: Single Case Prediction Result – Malignant




## Figure 4: Batch Prediction Results for Dataset




The system achieved high accuracy and demonstrated reliable performance across different prediction modes. The user-friendly interface and clear output messages make the system suitable for practical use while maintaining ethical considerations.